Sahaj Bhadja

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Summary

Graduate software engineer completing an MSCS at USC with 2+ years building **C++/Python systems**, concurrent pipelines, and financial applications; strong in data structures, algorithms, Linux, debugging, measurement-driven optimization, automated testing, and production delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Engineered **concurrent Python processing systems** with background workers, explicit state transitions, and failure isolation, executing 1,000+ jobs while improving throughput 3x under load and preventing cascading task failures.
- Measured and optimized **production application performance** across 5,000+ monthly workflows, using indexed access, validation, reconciliation, and diagnostics to reduce latency from 1.2 seconds to 350 milliseconds while maintaining 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed a **multithreaded Python processing engine** using ThreadPoolExecutor and fault-tolerant APIs, securely executing 1,000+ files per batch and maintaining stable performance across 10,000+ daily requests during peak load.
- Diagnosed complex processing failures across diverse inputs and model providers, implementing validation, persistent results, fallback handling, usage metrics, and automated tests while reducing turnaround time 60% through iterative performance improvements.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python simulation and evaluation pipelines** over 10,000 autonomous-driving episodes, combining map geometry and temporal signals to benchmark model behavior, quantify performance, and identify high-impact system improvements.
- Conducted diagnostic assessments across experimental runs, analyzing **metrics, constraints, and failure patterns** and communicating reproducible findings to researchers to prioritize configuration changes, accelerate root-cause investigation, and guide system refinement.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Optimized **object-oriented application and API components** using Singleton and Factory patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability.
- Owned cross-platform build and test automation across Windows, Linux, and macOS, using Jenkins, GitHub Actions, failure logs, coverage reports, and 20+ tests to deliver controlled production releases.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **performance-focused financial application services** for pricing, payments, refunds, disputes, and status updates across 50,000+ records, integrating Firebase transactions and validated webhooks while maintaining sub-200-millisecond API responses.
- Developed real-time transaction and payout workflows with idempotent processing, multi-currency support, and automated reconciliation, reducing manual payment handling 70% while improving booking confirmation speed 40% through Firebase synchronization.

Skills

Languages: C++, C, Python, Java, SQL, Bash, JavaScript, TypeScript
Computer Science: Data structures, algorithms, OOP, design patterns, concurrency, multithreading, synchronization
Systems: Linux, Windows, CPU scheduling, memory, I/O, distributed systems, real-time pipelines, system design
Performance: Profiling, benchmarking, diagnostics, metrics, root-cause analysis, latency optimization, high-throughput processing
Engineering: Git, Jenkins, GitHub Actions, CI/CD, Docker, PyTest, Mockito, automation, AI-assisted workflows, code review

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Systems Performance*
- Implemented and benchmarked **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

FinNexus Live | *Financial Systems, MongoDB, Kubernetes*
- Built a **scalable financial platform** supporting 50,000+ users and 1M+ transactions with secure bank linking, real-time synchronization, advanced filtering, token-based access, MongoDB persistence, and Kubernetes-based CI/CD automation.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Enhanced an **AI-assisted database application** across Java and TypeScript components, supporting 10+ SQL/NoSQL systems with schema exploration, query generation, reusable interfaces, Docker, Maven, and cross-platform automation workflows.